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**EDUCATION**

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**Stanford University**

Stanford, CA

B.S. **Mechanical Engineering**; Graduating in June 2027; GPA: **3.812**

Sep 2023 – Present

**Relevant coursework:** ENGR 14: Introduction to solid mechanics, ME 30: Thermodynamics, ME 70: Fluid dynamics, ME 80: Mechanics of Materials, ME 149: Mechanical Measurements, ME 102: Foundations of Product Realization, ENGR 40 M, ENGR 15: Dynamics, AA126: Aerospace Prototyping Lab

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**SKILLS**

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- **Programming & Controls:** Python, Arduino, MATLAB, JavaScript, C++
- **CAD & Simulation:** Fusion360, Onshape, SolidWorks, CNC, Luminary Cloud, ANSYS Fluent, HyperMesh, OptiStruct
- **Fabrication:** CNC machining, 3D printing, laser cutting, soldering, sand casting, welding

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**WORK EXPERIENCES**

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**Stanford Shape Lab**

Stanford, CA

*Undergraduate Research Assistant*

Dec 2023 – Present

Stanford Mechanical Engineering Department laboratory focused on the development of cutting-edge robotics, mechatronics, and sensing technologies with the primary goal to create dynamic 3D displays and haptic interfaces

<https://shape.stanford.edu/>

- Prototyped multiple wearable haptic-feedback devices, integrating LRAs, vibration motors, Dayton Audio exciters, DRV2605L drivers, amplifiers, and a TCA9548A I<sup>2</sup>C multiplexer into compact wristband designs.
- Worked through multi-channel hardware issues including power drops, signal noise, I<sup>2</sup>C address conflicts, motor placement, vibration strength, and wiring.
- Set up the Arduino and Node.js backend that connects the wristband to the web app, letting it send real-time vibration feedback based on handwriting errors.
- Ran hardware validation and pilot studies to measure latency, feedback precision, and user perception across prototype iterations.

**Stanford Solar Car Project**

Stanford, CA

*Member of Aero Sub team, Mechanical Division*

Mar 2025 – Present

Student-run organization at Stanford University dedicated to designing and building solar-powered cars including work on the most recent entry in the 2025 Formula Sun Grand Prix in Kentucky (2<sup>nd</sup> place)

- Integrated aerobody design with chassis, roll cage, and suspension hardpoints to meet structural, packaging, and safety constraints.
- Designed and surfaced aero body geometry with manufacturable, watertight CAD, improving curvature continuity, leading-edge shaping, and canopy integration.
- Ran aerodynamic simulations evaluating drag, separation behaviour, and crosswind/yaw stability.
- Built the CFD workflow for meshing, solver setup, and analysis of pressure distribution and wake structure.

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**PROJECTS**

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**Mechanical Design Project**

Apr – Jun 2025

*ME 102, Stanford University*

- Built a lever-actuated escapement and multi-stage gear train (planetary, bevel, worm-wheel). Performed kinematic, stress, and tolerance analyses. Fabricated via 3D printing in PLA and laser cutting.

**Mechanical Measurement Labs**

Apr – Jun 2025

*ME 149, Stanford University*

- Conducted rheology studies of non-Newtonian fluids, tensile testing of polymers, heat transfer experiments (Biot numbers, convection coefficients), and PIV vortex-ring analysis.

**Wing Test Project**

Sep – Nov 2025

*AA126, Stanford University*

- Designed and fabricated an airfoil wing section with custom mounting hardware, then calibrated the load-cell system and ran wind-tunnel tests across multiple angles of attack to analyze lift and drag performance.

**AA126: Wind Tunnel Project**

Sep 2025 – present

*AA126, Stanford University*

- Constructing a wind-tunnel assembly by CNC-cutting and heat-forming acrylic panels to build the contraction, test-section, and diffuser geometry, integrating a honeycomb flow straightener to condition the inlet stream and ensure uniform velocity profiles.